

Brac University
Department of Electrical & Electronic Engineering
Summer2026
Course Number : EEE309
Section :02



Assignment -01

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Section: 02

Semester:-Summer 2026

Answer to the Q. No-1

For, ~~n~~ p type Semiconductor,

zero bias $E = 0$

no excess carrier generation $g' = 0$.

Here,

spatial gradient is zero ; so the uniform spatial distribution, $\frac{\partial^2(\delta n)}{\partial x^2} = 0$

now,

The amb.

$$\frac{\partial(\delta p)}{\partial t} = D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E \frac{\partial(\delta p)}{\partial x} + g - \frac{\delta p}{\tau_p}$$

$$\Rightarrow \frac{\partial(\delta p)}{\partial t} = D_p(0) - \mu_p E(0) + 0 - \frac{\delta p}{\tau_p}$$

$$\Rightarrow \frac{\partial(\delta p)}{\partial t} = - \frac{\delta p}{\tau_p}$$

$$\Rightarrow \frac{\partial(\delta p)}{\partial t} + \frac{\delta p}{\tau_p} = 0$$

$$\Rightarrow s \delta p(s) - \delta p(0) + \frac{\delta p(s)}{\tau_{p0}} = 0$$

$$\Rightarrow \delta p(s) = \frac{\delta p(0)}{\left(s + \frac{1}{\tau_{p0}}\right)}$$

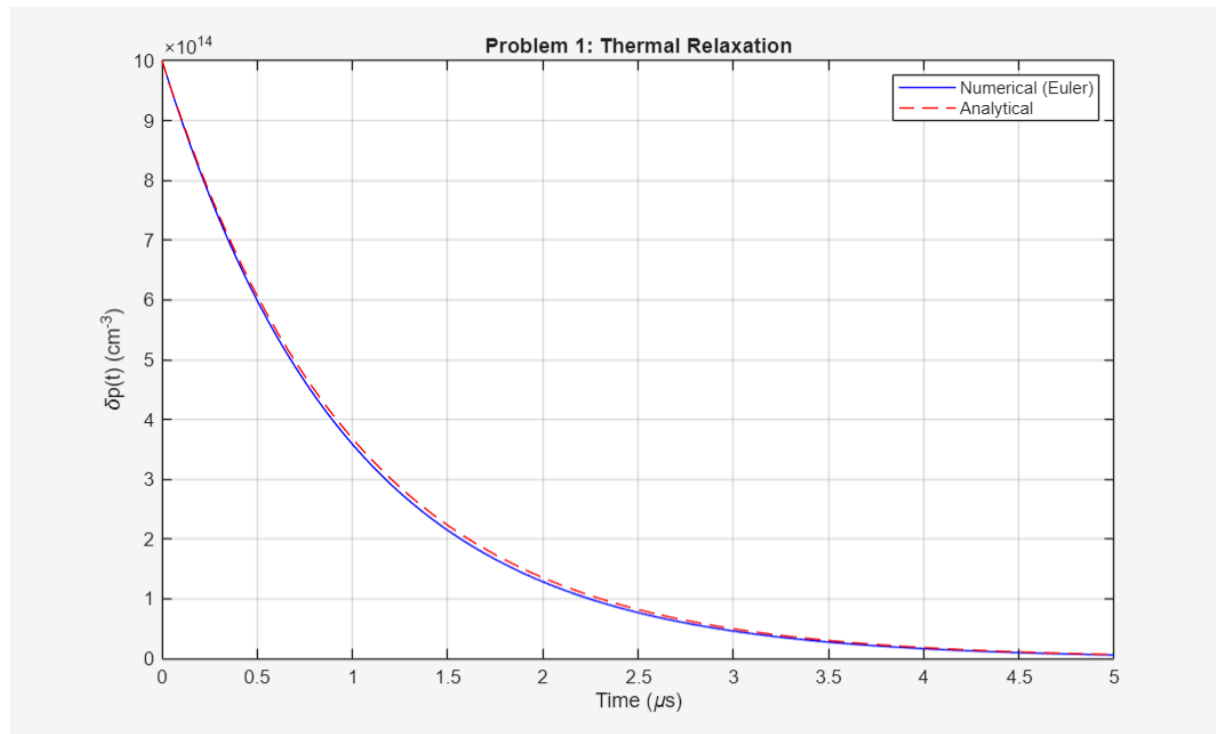
$$\therefore \delta p(t) = \delta p(0) e^{-\frac{t}{\tau_{p0}}}$$

Ans

MATLAB Code (Q1)

```
close all;
clear all;
clc;
tau_p = 1e-6;
dp0 = 1e15;
t = linspace(0, 5*tau_p, 100);
dt = t(2) - t(1);
dp_num = zeros(size(t)); % Numerical Euler Method
dp_num(1) = dp0;
for i = 1:length(t)-1
    dp_num(i+1) = dp_num(i) - dt * (dp_num(i) / tau_p);
end
dp_exact = dp0 * exp(-t / tau_p); % Exact Solution
figure;
plot(t*1e6, dp_num, 'b-', t*1e6, dp_exact, 'r--');
xlabel('Time (\mus)');
ylabel('\deltap(t) (cm^{-3})');
title('Problem 1: Thermal Relaxation');
legend('Numerical (Euler)', 'Analytical');
hold on;
grid on;
```

Code Simulation (Q1)



Answer to the Q: N-2

For p-type semiconductor,
constant generation g is constant
initial condition,
At $t=0$,

There are no excess carrier $\delta n(0) = 0$

$\therefore$ Uniform spatial distribution, $D_n \frac{\partial^2(\delta n)}{\partial x^2} = 0$

Now,

$$\frac{\partial(\delta n)}{\partial t} = D_n \frac{\partial^2(\delta n)}{\partial x^2} + \mu_n E \frac{\partial(\delta n)}{\partial x} + g = \frac{\delta n}{\tau_n}$$

$$\Rightarrow \frac{\partial(\delta n)}{\partial t} = g - \frac{\delta n}{\tau_n}$$

$$\Rightarrow \frac{\partial(\delta n)}{\partial t} + \frac{\delta n}{\tau_n} = g$$

$$\Rightarrow s \delta n(s) - \delta n(0) + \frac{\delta n(s)}{\tau_n} = \frac{g}{s}$$

$$\Rightarrow \delta n(s) = \frac{g}{s(s + \frac{1}{\tau_n})}$$

Now, Partial fraction and Inverse Laplace Transform.

$$\Rightarrow \delta n(s) = g \left(\frac{1}{s(s + \frac{1}{\tau_n})} \right) = g \left(\frac{1}{s} - \frac{1}{s + \frac{1}{\tau_n}} \right)$$

$$\Rightarrow \delta n(s) = g \tau_n \left(\frac{1}{s} - \frac{1}{s + \frac{1}{\tau_n}} \right)$$

$$\Rightarrow \delta n(s) = g \tau_n (1 - e^{-t/\tau_n})$$

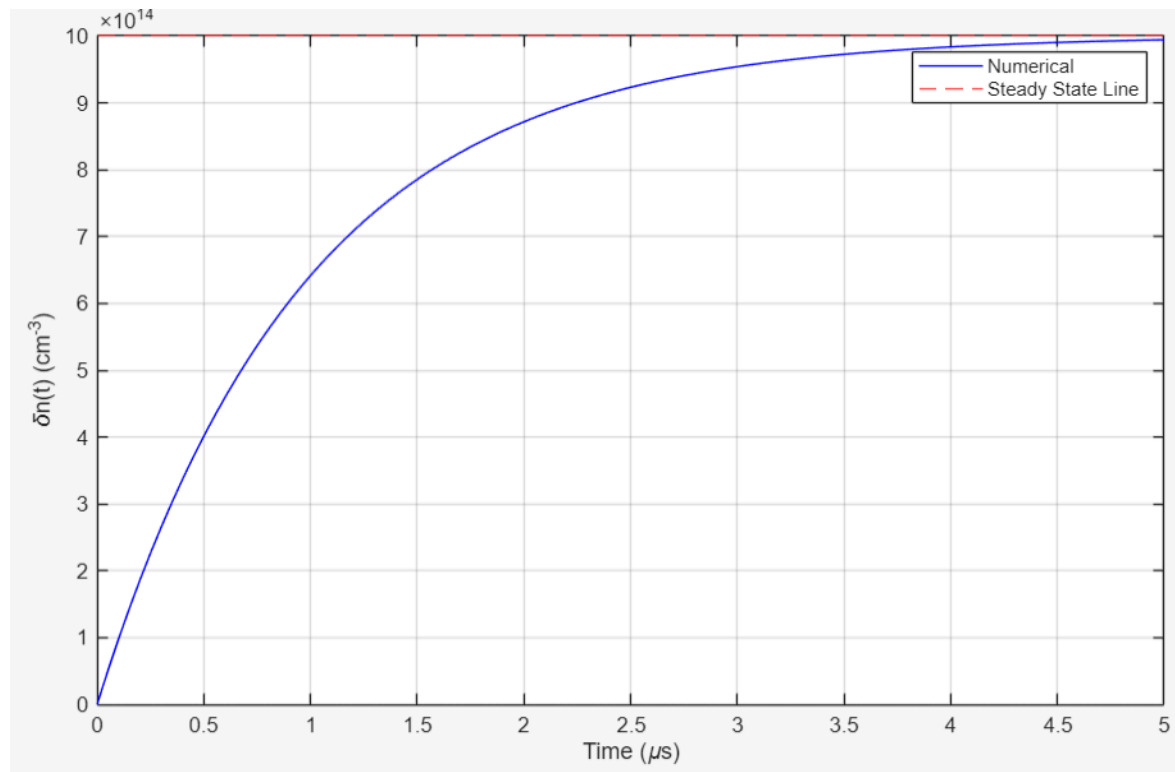
$$\delta n(s) = g \tau_n - g \tau_n e^{-t/\tau_n}$$

Ans

MATLAB Code (Q2)

```
close all;
clear all;
clc;
tau_p = 1e-6;
g = 1e21;
t = linspace(0, 5*tau_p, 100);
dt = t(2) - t(1);
dn = zeros(size(t));      % Euler Method
for i = 1:length(t)-1
    dn(i+1) = dn(i) + dt * (g-dn(i) / tau_p);
end
plot(t*1e6, dn, 'b-');
yline(g * tau_p, 'r--', 'Steady State');
xlabel('Time '); ylabel('\deltan(t)');
legend('Numerical', 'Steady State Line');
grid on;
hold on;
```

Code Simulation (Q2)



Ans to the Q. No-3

P type semiconductor,

zero bias, $E = 0$

no excess carrier $\delta n(0) = 0$

uniform spatial distribution $D_n \frac{\partial^2 (\delta n)}{\partial x^2} = g$

From Q-2,

$$\frac{\partial (\delta n)}{\partial t} = g - \frac{\delta n}{\tau_n}$$

now,

Phase-1

light ON ($0 \leq t \leq t'$)

$$g = g'$$

$$\Rightarrow \frac{\partial (\delta n)}{\partial t} = g' - \frac{\delta n}{\tau_n}$$

$$\Rightarrow S \cdot \delta n(S) - \delta n(0) + \frac{\delta n(S)}{\tau_n} = g'$$

$$\Rightarrow \delta n(t) = g' \tau_n (1 - e^{-t/\tau_n})$$

From (Q-2)
calculation,
Laplace and
inverse Laplace

Ans

again,

phase-2
light off ($t > t_1$)

here,

no excess generation $g=0$

$$\delta n(0) = \delta n(t_1) = g\tau_n(1 - e^{-t_1/\tau_n})$$

$$\therefore \frac{\partial(\delta n)}{\partial t} = g - \frac{\delta n}{\tau_n}$$

$$\Rightarrow \frac{\partial(\delta n)}{\partial t} = -\frac{\delta n}{\tau_n} \quad [g=0]$$

$$\Rightarrow \delta n(s) - \delta n(0) + \frac{\delta n(s)}{\tau_n} = 0$$

$$\Rightarrow \delta n(s) = \frac{\delta n(t_1)}{(s + \frac{1}{\tau_n})}$$

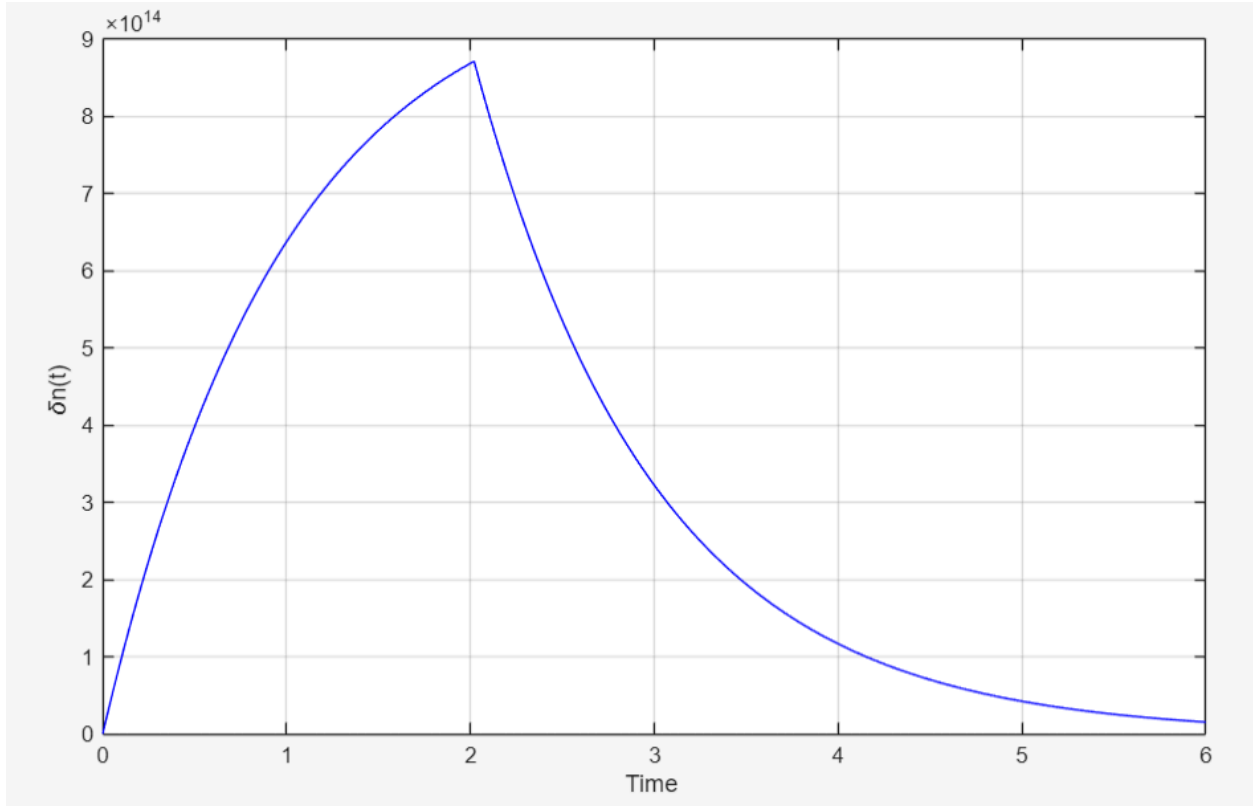
$$\Rightarrow \delta n(t) = \delta n(t_1) e^{-\left(\frac{t-t_1}{\tau_n}\right)}$$

$$\delta n(t) = g'\tau_n e^{-\frac{t-t_1}{\tau_n}} (1 - e^{-t/\tau_n}) \quad \underline{\text{Ans}}$$

MATLAB Code (Q3)

```
close all;
clear all;
clc;
tau_n = 1e-6;
g0 = 1e21;
t1 = 2*tau_n;
t = linspace(0, 6*tau_n, 200);
dt = t(2) - t(1);
dn = zeros(size(t));
for i = 1:length(t)-1
    g = (t(i) <= t1) * g0; % g0 when ON
    dn(i+1) = dn(i) + dt * (g - dn(i) / tau_n);
end
plot(t*1e6, dn, 'b-');
xlabel('Time ');
ylabel('\deltan(t) ');
grid on;
hold on;
```

Code Simulation (Q3)



Ans
Answer to the Q. No-4

given,
n-type semiconductor.

zero bias $E=0$

Steady state $\frac{d}{dt}=0$

surface illumination, at $x=0$.

now,

$$\frac{\partial(\delta p)}{\partial t} = D_p \frac{\partial^2(\delta p)}{\partial x^2} - \mu_p E \frac{\partial(\delta p)}{\partial x} + g - \frac{\delta p}{\tau_p}$$

$$\Rightarrow 0 = D_p \frac{\partial^2(\delta p)}{\partial x^2} - \frac{\delta p}{\tau_p}$$

$$\Rightarrow \frac{d^2(\delta p)}{dx^2} - \frac{\delta p}{D_p \tau_p} = 0$$

$$\Rightarrow \frac{\partial^2(\delta p)}{\partial x^2} - \frac{\delta p}{L_p^2} = 0 \quad [L_p^2 = D_p \cdot \tau_p]$$

Assume,

$$\frac{\partial^2(e^{mx})}{\partial x^2} - \frac{e^{mx}}{L_p^2} = 0$$

$$\Rightarrow m^2 e^{mx} - \frac{e^{mx}}{L_p^2} = 0$$

$$\Rightarrow m = \pm \frac{1}{L_p}$$

$$\therefore \delta p(x) = A e^{\frac{x}{L_p}} + B e^{-\frac{x}{L_p}}$$

now,

$$\lim_{x \rightarrow \infty} (A e^{\frac{x}{L_p}} + B e^{-\frac{x}{L_p}}) = 0$$

$$\Rightarrow \delta p(x) = B e^{-\frac{x}{L_p}} \quad [x \rightarrow \infty]$$

$$\Rightarrow \delta p(0) = B e^{-0}$$

$$\Rightarrow B = \delta p_s$$

$$\delta p(x) = \delta p_s e^{-\frac{x}{L_p}} \quad \underline{\text{Ans}}$$

MATLAB Code (Q4)

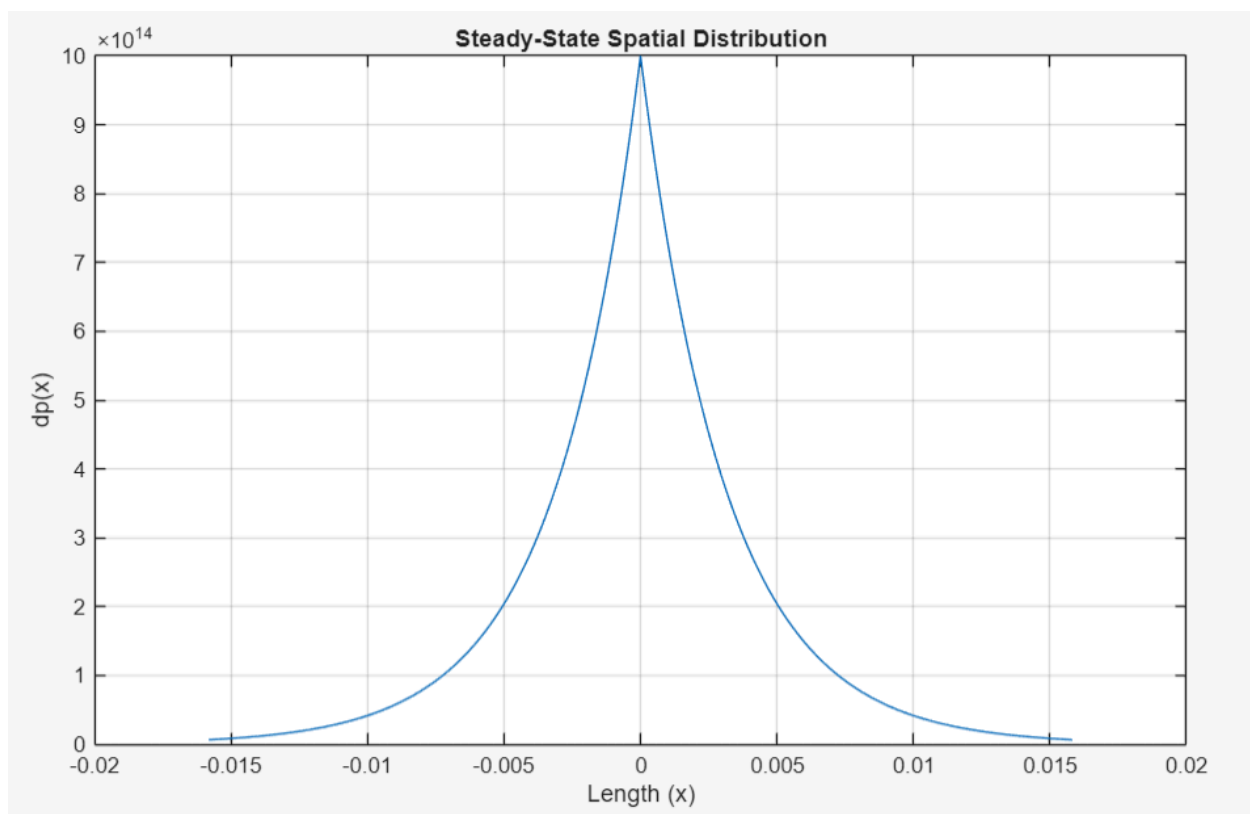
```
clear all;
close all;
clc;
Dp = 10;
tau = 1e-6;
Lp = sqrt(Dp*tau);
p0 = 1e15;
h = 0.01*Lp;
% Positive side
x_pos = 0:h:5*Lp;
p_pos = zeros(1,length(x_pos));
q_pos = zeros(1,length(x_pos));
p_pos(1) = p0;
q_pos(1) = -p0/Lp;
% Euler Method - Positive side
for k = 2:length(x_pos)
    p_pos(k) = p_pos(k-1) + h*q_pos(k-1);
    q_pos(k) = q_pos(k-1) + h*(p_pos(k-1)/(Dp*tau));
end
% Negative side - symmetric
x_neg = -fliplr(x_pos(2:end));
p_neg = fliplr(p_pos(2:end));
q_neg = -fliplr(q_pos(2:end));
% Combine negative and positive sides
x = [x_neg, x_pos];
```

```

p = [p_neg, p_pos];
q = [q_neg, q_pos];
% Analytical solution
p_exact = p0*exp(-abs(x)/Lp);
figure
plot(x,p)
grid on
xlabel('Length (x)')
ylabel('dp(x)')
title('Steady-State Spatial Distribution')

```

Code Simulation (Q4)



Answer to the Q-5

given,

no excess generation $g=0$

Transient with diffusion and recombination.

$$\frac{\partial(\delta p)}{\partial t} = D_p \frac{\partial^2(\delta p)}{\partial x^2} - \frac{\delta p}{\tau_p} - \mu_p E \frac{\partial(\delta p)}{\partial x} \quad \text{--- (i)}$$

Here,

$$S_p(x, t) = P'(x, t) e^{-t/\tau_p}$$

now,

Differentiating $S_p(x, t)$

$$\begin{aligned} \frac{\partial(S_p)}{\partial t} &= \frac{\partial P'}{\partial t} e^{-t/\tau_p} - \frac{1}{\tau_p} P'(x, t) e^{-t/\tau_p} \\ &= e^{-t/\tau_p} \cdot \frac{\partial P'}{\partial t} - \frac{\delta p}{\tau_p} \quad \text{--- (ii)} \end{aligned}$$

(i) $\Rightarrow$

$$\Rightarrow e^{-t/\tau_p} \frac{\partial P'}{\partial t} - \frac{\delta p}{\tau_p} = D_p e^{-t/\tau_p} \frac{\partial^2 P'}{\partial x^2} - \mu_p E e^{-t/\tau_p} \frac{\partial P'}{\partial x}$$

$$\Rightarrow e^{-t/\tau_p} = D_p e^{-t/\tau_p} \frac{\partial^2 P'}{\partial x^2} - \mu_p E e^{-t/\tau_p} \frac{\partial P'}{\partial x}$$

$$\Rightarrow \frac{\partial P'(x, t)}{\partial t} = D_p \frac{\partial^2 P'(x, t)}{\partial x^2} - \mu_p E \frac{\partial P'(x, t)}{\partial x}$$

$$\therefore P'(x, t) = \frac{1}{\sqrt{4\pi D_p t}} \exp \left[-\frac{(x - \mu_p E_0 t)^2}{4 D_p t} \right]$$

$$S_p(x, t) = \frac{e^{-t/\tau_p}}{\sqrt{4\pi D_p t}} \exp \left[-\frac{(x - \mu_p E_0 t)^2}{4 D_p t} \right]$$

MATLAB Code (Q5)

```
clear all;
close all;
clc;
Dp = 10;
tau = 1e-6;
Lp = sqrt(Dp*tau);
p0 = 1e15;
h = 0.01*Lp;
x_pos = 0:h:5*Lp;
dt = 0.4*h^2/Dp;
t = 0:dt:5*tau;
N = length(x_pos);
M = length(t);
% Initial condition
p_pos = zeros(M,N);
% Boundary conditions
p_pos(:,1) = p0;
p_pos(:,N) = p0*exp(-5);
% Euler Method
for n = 1:M-1
    for k = 2:N-1
        p_pos(n+1,k) = p_pos(n,k) ...
            + dt*(Dp*(p_pos(n,k+1)-2*p_pos(n,k)+p_pos(n,k-1))/h^2 ...
            - p_pos(n,k)/tau);
    end
end
end
```



```

% Negative side - symmetric
x_neg = -fliplr(x_pos(2:end));
p_neg = fliplr(p_pos(:,2:end));
% Combine both sides
x = [x_neg x_pos];
p = [p_neg p_pos];
% Steady-state solution from Problem 4
p_exact = p0*exp(-abs(x)/Lp);
% 2D surface plot
figure(1)
surf(x,t,p)
shading interp
grid on
xlabel('Length (x)')
ylabel('Time (s)')
zlabel('dp(x,t)')
title('2D surface/color plot of dp(x, t)')
% dp vs x at different times
figure(2)
hold on
plot(x,p(find(t>=0.5*tau,1),:))
plot(x,p(find(t>=tau,1),:))
plot(x,p(find(t>=2*tau,1),:))
plot(x,p(find(t>=5*tau,1),:))
plot(x,p_exact,'--')
grid on
xlabel('Length (x)')

```

```
ylabel('dp(x,t)')
legend('0.5\tau', '\tau', '2\tau', '5\tau', 'Steady State')
title('Spatial Distribution')

% dp vs t at different positions

figure(3)
hold on
plot(t,p(:,find(x==0,1)))
plot(t,p(:,find(abs(x-Lp)==min(abs(x-Lp)),1)))
plot(t,p(:,find(abs(x-2*Lp)==min(abs(x-2*Lp)),1)))
grid on
xlabel('Time (s)')
ylabel('dp(x,t)')
title('Time Dependence')
```